

PAPER_685: UQFF Implications for Primordial Black Holes as Dark Matter

Author: Daniel Murphy

Framework: Unified Quantum Field Framework (UQFF) v5.30

Session: 173

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Class: UQFFPBHDarkMatterImplications | CP4 #269

Source: grok_share_fc21e30c24b4.txt

Domain: Dark Matter / PBH Cosmology

Abstract

The UQFF extended PBH lifetime shifts the critical survival mass from $M_{\text{crit,std}} \approx 5 \times 10^{11}$ kg to $M_{\text{crit,UQFF}} = M_{\text{crit,std}} / \tau_{\text{ratio}}^{1/3} \approx 0.46 M_{\text{crit,std}}$, expanding the viable PBH dark matter mass window. The DM fraction is boosted: $f_{\text{PBH,UQFF}} = f_{\text{PBH,GR}} \cdot \tau_{\text{ratio}}^{2/3}$. In the UQFF framework the mass window 10^{10} - 10^{17} kg is fully open as dark matter.

Primary UQFF Equation

$$M_{\text{crit,UQFF}} = \frac{M_{\text{crit,std}}}{\tau_{\text{ratio}}^{1/3}}, \quad f_{\text{PBH,UQFF}} = f_{\text{PBH,GR}} \cdot \tau_{\text{ratio}}^{2/3}$$

Parameters

Parameter	Value	Description
$M_{\text{crit,std}}$	5×10^{11} kg	Standard survival mass
τ_{ratio}	~ 11	UQFF lifetime boost
DM window	10^{10} - 10^{17} kg	UQFF viable range

UQFF Constants

Constant	Value	Description
ρ_{UA}	7.09×10^{-36} kg/m ³	Universal Aether density
ρ_{SCm}	7.09×10^{-37} kg/m ³	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	5×10^{-4} day ⁻¹	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	3.38×10^{23} J·m	Magnetic string coupling
γ	5×10^{-5} / 86400 s ⁻¹	Aether oscillation decay

C++ Implementation

```
#include "UQFFPBHDarkMatterImplications.h"

UQFFPBHDarkMatterImplications obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import UQFFPBHDarkMatterImplicationsCalculator

calc = UQFFPBHDarkMatterImplicationsCalculator()
result = calc.compute() # returns all UQFF quantities
print(result)
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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

Carr & Hawking 1974; Carr et al. 2021 (PBH DM); Raidal et al. 2019; UQFF Session 173

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